

ASSIGNMENT 6 (Structural Estimation of Auctions)

Overview. This assignment uses classroom experiment data to study nonparametric and structural estimation of first-price auctions. The three questions build on each other: (1) nonparametric identification under correct specification, (2) observational equivalence and testing, and (3) parametric structural estimation when the nonparametric approach fails. A bonus question tests the strategic equivalence between sealed-bid and Dutch auction formats.

Instructions. Work individually or in pairs. Submit a self-contained report with documented code (Jupyter notebook or appendix). Use classroom experiment data where available; otherwise use simulated data from the provided template.

Setup. Throughout, n risk-neutral bidders compete for a single indivisible object. The experiment uses $n = 5$ and $R = 10$ rounds per treatment.

Treatment 1: FPSB Procurement (IPV). A buyer solicits sealed bids from n suppliers with private costs $c_i \stackrel{\text{i.i.d.}}{\sim} F$ on $[\underline{c}, \bar{c}]$. The lowest bidder wins and is paid her bid; profit is $b_i - c_i$. In the experiment, $c_i \sim \text{Uniform}[200, 400]$.

Treatments 2 & 3: FPSB and Dutch (CV). A common value $V \sim \text{Uniform}[500, 1000]$ with private signals $s_i = V + \varepsilon_i$, $\varepsilon_i \stackrel{\text{i.i.d.}}{\sim} \text{Uniform}[-100, 100]$. The highest bidder wins and pays her bid; payoff is $V - b_i$. Each CV round was conducted in both sealed-bid (Treatment 2) and Dutch descending-clock (Treatment 3) formats, providing within-round paired observations.

	IPV (Procurement)	CV (Common Value)
FPSB	Treatment 1	Treatment 2
Dutch	—	Treatment 3

Question 1: Nonparametric Cost Recovery under IPV.

Goal. Adapt the GPV (Guerre, Perrigne, Vuong 2000) nonparametric inversion to the procurement setting and apply it to Treatment 1 data. The key difference from a standard (highest-bid-wins) auction is that procurement bidders shade *upward*: derive the appropriate first-order condition and resulting inversion formula, then implement it.

- (a) Derive the symmetric Bayes–Nash equilibrium first-order condition for the procurement auction. Show that the inversion mapping from bids to costs takes the form

$$c_i = b_i - \frac{1 - G(b_i)}{(n-1)g(b_i)},$$

where G and g are the CDF and density of equilibrium bids. Explain the economics of upward shading.

- (b) Implement the inversion using kernel density estimation of G and g . Since the experiment provides both bids and true costs, evaluate how well the method recovers the latent cost distribution (plots and error metrics).
- (c) Derive the closed-form equilibrium bid function $\beta(c)$ for $c_i \sim \text{Uniform}[\underline{c}, \bar{c}]$, and verify that the inversion recovers c analytically. Compare $\beta(c)$ to observed bidding behavior.
- (d) State the Revenue Equivalence Theorem for procurement. Compute the theoretical expected buyer payment under FPSB and second-price procurement, and compare with observed payments in Treatment 1.

Question 2: Observational Equivalence and Testing.

Goal. Demonstrate that an econometrician who observes only bids cannot distinguish IPV from CV when n is fixed (Laffont and Vuong, 1996), and then show how variation in the number of bidders breaks this equivalence (Haile, Hong, and Shum, 2003).

- (a) Apply the GPV inversion to *both* FPSB datasets — the procurement inversion for Treatment 1 and the standard inversion $v_i = b_i + G(b_i)/[(n-1)g(b_i)]$ for Treatment 2. Can you tell from the recovered pseudo-values alone which dataset is IPV and which is CV?
- (b) Now use the observed true costs (Treatment 1) and common value/signals (Treatment 2) to evaluate the accuracy of the pseudo-values. In the CV case, characterize what the pseudo-values are actually estimating and explain the source of the bias (relate to the winner's curse).
- (c) Relate parts (a)–(b) to the observational equivalence result in Laffont and Vuong (1996). What source of variation could break this equivalence?
- (d) Using simulated data with $n \in \{3, 5, 7, 10\}$, test whether the recovered pseudo-value distribution depends on n , separately for IPV and CV. Explain the difference and propose a formal test following Haile, Hong, and Shum (2003).

Question 3: Structural Estimation of the Common Value Model.

Goal. When the data-generating process is known to be CV, the nonparametric GPV approach no longer applies. This question asks you to estimate the structural parameters $\theta = (\omega, \delta)$ of the CV model $V \sim \text{Uniform}[0, \omega]$, $\varepsilon_i \sim \text{Uniform}[-\delta, \delta]$ using both signal data and bid data alone. (For the experiment, work with the shifted value $V' = V - 500$ so that $\omega = 500$, or estimate ω directly.)

- (a) **Estimation with signals.** Derive $\text{Var}(s_i)$ and $\text{Cov}(s_i, s_j)$ as functions of (ω, δ) . Use these two moment conditions to construct a method-of-moments estimator and report estimates.
- (b) **Estimation from bids only.** Implement a simulated method of moments (SMM) estimator: for candidate (ω, δ) , compute the equilibrium bid function, simulate auctions, and match moments of the simulated and observed bid distributions. The equilibrium

for this model is characterized by Wilson (1977); see Paarsch and Hong (2006, Ch. 4) for the derivation and Laffont, Ossard, and Vuong (1995) for the SMM methodology.

- (c) Compare the estimates and efficiency of (a) and (b). Which is more practical when signals are unobserved?
- (d) **Misspecification bias.** Using your SMM simulator, generate CV auction data and apply the IPV-GPV inversion. Characterize the resulting pseudo-value distribution and plot the bias as a function of δ/ω . What happens as $\delta \rightarrow 0$ and as δ grows large?

References.

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